



Board-Review Quality • Comprehensive Academic Reference

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2025 Edition

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Chapter 1 Definition and Overview

Definition and Overview

Pruritus is defined as an ****unpleasant cutaneous sensation that provokes the desire to scratch or rub the affected area****. It is fundamentally distinct from pain, though both share overlapping neural pathways. Pruritus can be localized or generalized, acute or chronic, and may occur with or without visible skin changes. Chronic pruritus is profoundly debilitating — disrupting sleep, impairing concentration, and contributing to clinical depression and social withdrawal. Patients with chronic itch report a quality-of-life burden comparable to chronic pain disorders.

Classification of Pruritus

By Distribution:

- Localized — confined to a specific anatomical region
- Generalized — widespread, affecting multiple body areas

By Skin Findings:

- Pruritus *with* skin changes (primary dermatologic conditions)
- Pruritus *with normal-appearing skin* (suggesting systemic disease)

By Duration:

- Acute — less than 6 weeks
- Chronic — 6 weeks or longer

KEY TEACHING POINTS

Pruritus with normal skin is a RED FLAG for underlying systemic disease — always investigate.

Chronic pruritus is defined as lasting 6 weeks or longer.

The itch-scratch cycle is self-perpetuating: scratching causes barrier damage, which releases more pruritogens, which causes more itching.

Pathogenesis: The Itch-Scratch Cycle

When skin contacts allergens or irritating substances, epidermal nerve endings (unmyelinated C-fibers and thinly myelinated A-delta nociceptive fibers) become irritated and inflamed. This triggers mast cell degranulation with release of histamine, serotonin, substance P, prostaglandins, leukotrienes, and cytokines such as IL-31.

Histamine binds H1 receptors on sensory nerve endings, generating the itch sensation. The patient scratches, causing mechanical skin barrier damage and a secondary inflammatory response with additional pruritogen release, intensifying the itch. This self-perpetuating loop is the **itch-scratch cycle**.

Key Mediators of Itch:

- Histamine (H1 and H4 receptors)
- Serotonin
- Substance P
- Prostaglandins and leukotrienes
- Interleukin-31 (IL-31)
- Protease-activated receptors (PAR-2)

MCQ TRAPS & PITFALLS

IL-31 is a cytokine specifically implicated in pruritus, not just general inflammation.

Unmyelinated C-fibers mediate itch — myelinated A-beta fibers mediate touch, not itch.

Histamine is NOT the only mediator — many forms of itch are histamine-independent, such as uremic pruritus and cholestatic pruritus.

Complications of Pruritus

Chronic scratching leads to significant morbidity affecting both physical and psychological health.

Cutaneous Complications:

- Excoriations, erosions, and ulceration from repetitive trauma
- Secondary bacterial infection (impetigo, cellulitis)
- Eczema herpeticum — HSV dissemination across eczematous skin
- Fungal superinfection
- Lichenification — thickened, leathery skin with accentuated skin markings
- Post-inflammatory hyperpigmentation or hypopigmentation
- Scarring and textural changes

Psychological and Functional Impact:

- Sleep disruption and insomnia
- Anxiety and depression
- Social withdrawal and embarrassment
- Impaired concentration and work performance
- Reduced health-related quality of life

IMPORTANT

Eczema herpeticum (Kaposi varicelliform eruption) is a medical emergency requiring urgent antiviral therapy.

Lichenification is a hallmark of chronic scratching and indicates long-standing disease.

Assess all patients with chronic pruritus for psychological comorbidity — screen for depression.

Pruritus WITH Skin Changes

When pruritus is accompanied by visible skin lesions, the diagnostic approach focuses on identifying the primary dermatologic condition. Morphology, distribution, and character of the skin changes provide critical diagnostic clues. Assessment requires detailed history including onset, duration, triggering factors, occupational exposures, travel history, medication review, and associated systemic symptoms.

Inflammatory Dermatoses

Eczema (Atopic Dermatitis)

Erythematous, scaly, often excoriated patches with intense paroxysmal pruritus. Distribution is age-dependent: face and extensor surfaces in infants and toddlers; flexural areas (antecubital fossae, popliteal fossae, neck, wrists, ankles) in older children and adults. Associated with the atopic triad: eczema, allergic rhinitis, and asthma. Elevated serum IgE in many patients.

Psoriasis

Well-demarcated erythematous plaques with silky, silvery-white scale. Common sites include extensor surfaces (elbows, knees), scalp, lumbosacral region, and nails. Auspitz sign: pinpoint bleeding when scales are scraped off. Nail findings include pitting, onycholysis, and oil-drop discoloration. Generally less pruritic than eczema, though significant itching occurs in intertriginous and genital disease. Associated with psoriatic arthritis in up to 30% of patients.

Lichen Planus

Polygonal, purple (violaceous), pruritic, flat-topped (planar) papules and plaques. Characteristic fine white lines called Wickham striae on the surface. Commonly affects wrists, ankles, lower back, oral mucosa, and genitalia. Koebner phenomenon: new lesions develop at sites of trauma. Oral lichen planus may present with erosive, painful patches on buccal mucosa.

Dermatitis Herpetiformis

Gluten-associated dermatosis presenting as excoriated papulovesicular eruption on elbows, knees, buttocks, and scalp. Patients often scratch lesions before they can be examined, so primary vesicles may be absent. Strongly associated with celiac disease — patients may have GI symptoms (bloating, diarrhea, malabsorption) or may be asymptomatic. Immunofluorescence shows granular IgA deposits at dermal papillae. Treatment: gluten-free diet and dapsone.

Inflammatory Dermatoses Causing Localized Pruritus

Condition	Key Morphology	Distribution	Special Features
Eczema (Atopic Dermatitis)	Erythematous, scaly, excoriated patches	Age-dependent: face/extensors (infants); flexures (adults)	Intense paroxysmal itch; associated with atopic triad; elevated IgE
Psoriasis	Well-demarcated plaques with silvery-white scale	Extensors (elbows, knees), scalp, lumbosacral	Auspitz sign; nail pitting; psoriatic arthritis in 30%
Lichen Planus	Polygonal, violaceous, flat-topped papules	Wrists, ankles, lower back, oral mucosa	Wickham striae; Koebner phenomenon
Dermatitis Herpetiformis	Excoriated papulovesicles	Elbows, knees, buttocks, scalp	Associated with celiac disease; IgA at dermal papillae; treat with dapsone and gluten-free diet

Contact Dermatitis

Allergic Contact Dermatitis

Delayed-type (Type IV) hypersensitivity reaction to an allergen. Presents with erythema, vesicles, and edema at the site of contact. Common allergens include nickel (jewelry, belt buckles), fragrances, preservatives (parabens, formaldehyde), neomycin, poison ivy/oak (urushiol), and rubber accelerators. The eruption typically appears 48-72 hours after exposure.

Irritant Contact Dermatitis

Non-immunologic damage to the skin from direct chemical or physical irritation. Most common type of contact dermatitis. Common irritants include soaps, detergents, solvents, acids, alkalis, and frequent hand washing. Hands are most commonly affected, especially in healthcare workers and cleaners.

KEY TEACHING POINTS

Patch testing is the gold standard for identifying the causative allergen in allergic contact dermatitis.

Always ask about occupation, hobbies, and personal care products when evaluating contact dermatitis.

Poison ivy dermatitis presents with linear streaks of vesicles — the pattern reflects brushing against the plant.

Allergic vs. Irritant Contact Dermatitis

Feature	Allergic Contact Dermatitis	Irritant Contact Dermatitis
Mechanism	Type IV hypersensitivity	Direct chemical/immunologic damage
Onset after exposure	48-72 hours	Minutes to hours
Dose-dependent	No (sensitization required)	Yes
Patch testing	Positive	Negative
Common allergens/irritants	Nickel, fragrances, urushiol	Soaps, detergents, solvents

Infectious Causes of Localized Pruritus

Tinea (Dermatophytosis)

Fungal infection caused by dermatophytes (Trichophyton, Microsporum, Epidermophyton). Presents as annular, scaly, erythematous plaques with central clearing and an active, raised, advancing border. Distribution defines the type: tinea corporis (body), tinea cruris (groin), tinea pedis (feet), tinea capitis (scalp), tinea unguium/onychomycosis (nails). Diagnosis: KOH preparation showing hyphae. Topical antifungals (terbinafine, clotrimazole) for localized disease; oral terbinafine or itraconazole for extensive or nail involvement.

Scabies

Caused by the mite *Sarcoptes scabiei*. Intense pruritus that is characteristically worse at night. Burrows (thin, serpentine, gray-white tracks) are the pathognomonic lesion, found in web spaces of fingers, wrists, elbows, axillae, waistline, and genitalia. In crusted (Norwegian) scabies, thousands of mites are present in thick, crusted scales — highly contagious and seen in immunocompromised patients. Treatment: permethrin 5% cream applied to entire body from neck down, wash off after 8-14 hours; treat all household contacts simultaneously.

Head Lice (Pediculosis Capitis)

Intense scalp pruritus with nits (eggs) firmly attached to hair shafts. Posterior cervical lymphadenopathy may be present. Transmission through direct head-to-head contact. Treatment: permethrin 1% lotion or malathion 0.5% lotion; remove nits with fine-toothed comb.

Varicella-Zoster Reactivation (Herpes Zoster)

Reactivation of latent VZV in dorsal root or cranial nerve ganglia. Prodromal pain and pruritus in a dermatomal distribution, followed by grouped vesicles on an erythematous base. Painful more than pruritic, but itching is common. Complications include post-herpetic neuralgia, especially in patients over 60 years. Treatment: acyclovir, valacyclovir, or famciclovir within 72 hours of rash onset.

OSCE PEARLS

Ask about nocturnal worsening of itch — this is a hallmark of scabies.

Always examine the finger webs and wrist creases when scabies is suspected.

Crusted scabies can mimic psoriasis or eczema — consider it in any immunocompromised patient with widespread scaling.

Other Dermatologic Conditions

Urticaria (Hives)

Transient, erythematous, edematous wheals with central pallor. Individual lesions typically last less than 24 hours, migrating and resolving without scarring. Caused by mast cell degranulation with histamine release. Acute urticaria (less than 6 weeks) is often triggered by infections, foods, or medications. Chronic urticaria (more than 6 weeks) is frequently idiopathic or autoimmune. Treatment: second-generation H1 antihistamines (cetirizine, loratadine, fexofenadine) as first-line; up-dosing up to 4x standard dose for chronic urticaria; add H2 blockers or omalizumab for refractory cases.

Drug Eruptions

Drug-induced pruritus may occur with or without visible rash. Common culprits include antibiotics (penicillins, sulfonamides), NSAIDs, opioids, and ACE inhibitors. Morphology ranges from maculopapular exanthem to urticaria to severe reactions such as Stevens-Johnson syndrome or drug reaction with eosinophilia and systemic symptoms (DRESS). Management: discontinue the offending agent; antihistamines for symptomatic relief.

Nodular Prurigo

Characterized by intensely pruritic, lichenified nodules on extensor surfaces of limbs. Represents end-stage change from chronic scratching/rubbing. Often associated with atopic dermatitis, but can occur independently. Treatment is challenging — potent topical corticosteroids under occlusion, intralesional triamcinolone, phototherapy, and thalidomide in refractory cases.

Summary of Pruritic Conditions with Skin Changes

Category	Condition	Key Diagnostic Feature
Inflammatory	Eczema	Age-dependent distribution; atopic triad
Inflammatory	Psoriasis	Well-demarcated plaques; Auspitz sign
Inflammatory	Lichen planus	Wickham striae; violaceous papules
Inflammatory	Dermatitis herpetiformis	IgA deposits; celiac association
Contact	Allergic contact dermatitis	Delayed onset; patch test positive
Contact	Irritant contact dermatitis	Dose-dependent; common in hands
Infectious	Tinea	Annular plaques with central clearing; KOH positive
Infectious	Scabies	Burrows; nocturnal itch; contact history
Infectious	Herpes zoster	Dermatomal grouped vesicles

Category	Condition	Key Diagnostic Feature
Other	Urticaria	Transient wheals; individual lesions < 24 hours
Other	Nodular prurigo	Lichenified nodules on extensors

Pruritus WITHOUT Skin Changes

Generalized pruritus with normal-appearing skin is a critical clinical scenario demanding thorough systemic investigation. The absence of primary skin lesions suggests internal disease, hematologic malignancy, metabolic derangement, or psychogenic etiology. Excoriations from scratching are secondary and should not be considered the cause of the itch.

Systemic Causes of Generalized Pruritus

Hematologic Malignancies

- **Hodgkin lymphoma:** Generalized pruritus is a classic early symptom, occurring in up to 30% of patients. May precede diagnosis by months. Characteristically worse after alcohol ingestion (alcohol-induced pain in affected lymph nodes).
- **Polycythemia vera:** Aquagenic pruritus — intense itching triggered by contact with water, especially after a warm bath. Due to abnormal mast cell degranulation.
- **Myelodysplastic syndromes and leukemia:** Pruritus may occur as a paraneoplastic phenomenon.

Hepatobiliary Disease

- **Cholestatic liver disease:** Primary biliary cholangitis, primary sclerosing cholangitis, obstructive jaundice (gallstones, pancreatic carcinoma). Bile salts deposit in the skin and stimulate peripheral itch receptors. Pruritus is often most severe on palms, soles, and pressure areas.
- **Hepatitis C virus:** Pruritus may be a presenting symptom even before liver function derangement.

Renal Disease

- **Uremic pruritus:** Affects 40-90% of patients on hemodialysis. Mechanism involves uremic toxins, opioid receptor imbalance, and microinflammation. Typically generalized but may be localized to the back. Often refractory to antihistamines.

Endocrine Disorders

- **Hyperthyroidism:** Increased skin blood flow and temperature increase histamine release.
- **Hypothyroidism:** Dry skin (xerosis) with secondary pruritus.
- **Diabetes mellitus:** Pruritus may result from xerosis, candidal infection, or neuropathy-associated itch.

Solid Malignancies

Paraneoplastic pruritus is associated with lung, breast, gastric, and colorectal carcinomas. It results from cytokine and mediator release by tumor cells. Pruritus that is unexplained and progressive in an older patient should raise suspicion for occult malignancy.

IMPORTANT

Progressive unexplained pruritus in an elderly patient should prompt a thorough malignancy workup, including imaging and blood tests.

Aquagenic pruritus (itching triggered by water contact) is highly specific for polycythemia vera.

Hodgkin lymphoma pruritus characteristically worsens after alcohol consumption.

Systemic Causes of Generalized Pruritus WITHOUT Skin Changes

System	Condition	Key Features	Diagnostic Tests
Hematologic	Hodgkin lymphoma	Worse after alcohol; night sweats	LDH, CT chest/abdomen, lymph node biopsy
Hematologic	Polycythemia vera	Aquagenic pruritus after warm bath	JAK2 mutation, CBC, erythropoietin level
Hematologic	Myelodysplastic syndromes	Fatigue, cytopenias	Peripheral smear, bone marrow biopsy
Hepatobiliary	Primary biliary cholangitis	Middle-aged women; pruritus on palms/soles	Anti-mitochondrial antibody (AMA), ALP, GGT
Hepatobiliary	Cholestatic jaundice	Pale stools, dark urine	LFTs, abdominal ultrasound, MRCP
Hepatobiliary	Hepatitis C	Risk factors: IV drug use, transfusions	Hepatitis C antibody, viral load
Renal	Uremic pruritus	Hemodialysis patients; generalized	BUN, creatinine, eGFR
Renal	Chronic kidney disease	Multifactorial: uremic toxins, opioid imbalance	Renal function tests
Endocrine	Hyperthyroidism	Warm skin, weight loss, tachycardia	TSH, free T4
Endocrine	Hypothyroidism	Xerosis, fatigue, cold intolerance	TSH, free T4
Endocrine	Diabetes mellitus	Xerosis, candidal intertrigo	HbA1c, fasting glucose
Malignancy	Solid tumors	Paraneoplastic; older patient	Age-appropriate cancer screening, CT imaging

Diagnostic Approach to Pruritus

A systematic, stepwise approach is essential for evaluating pruritus, particularly when no primary skin lesion is identified.

History

- Duration: acute versus chronic (chronic defined as 6 weeks or more)
- Distribution: localized versus generalized
- Timing: nocturnal worsening suggests scabies
- Triggers: water contact (polycythemia vera), alcohol (Hodgkin lymphoma), heat, stress

- Associated symptoms: weight loss, night sweats, fever (B-symptoms suggesting malignancy)
- Medication history: opioids, antibiotics, NSAIDs, ACE inhibitors
- Past medical history: atopy, liver disease, renal disease, thyroid disease, diabetes
- Travel history and contact history: scabies, tropical infections
- Family history: atopy, dermatitis herpetiformis (celiac disease)
- Psychological history: anxiety, depression, obsessive-compulsive disorder, delusions of parasitosis

Examination

Skin Assessment:

- Document presence or absence of primary skin lesions
- Excoriations (secondary to scratching, NOT primary pathology)
- Lichenification: chronicity marker
- Xerosis: dry skin contributing factor
- Distribution pattern: may suggest specific diagnosis

Systemic Examination:

- Lymphadenopathy: lymphoma, infection
- Hepatosplenomegaly: hematologic malignancy, liver disease
- Thyroid enlargement: thyroid dysfunction
- Signs of chronic liver disease: jaundice, spider nevi, palmar erythema
- Signs of chronic renal disease: pallor, uremic frost

Investigations

Baseline Blood Tests:

- CBC with differential (eosinophilia in allergy/parasites; polycythemia; cytopenias)
- Liver function tests (cholestatic pattern: elevated ALP, GGT, bilirubin)
- Renal function tests (BUN, creatinine, eGFR)
- Thyroid function tests (TSH, free T4)
- Fasting glucose and HbA1c
- Iron studies (iron deficiency can cause generalized pruritus)
- ESR and CRP (inflammatory markers)

Targeted Investigations Based on Clinical Suspicion:

- Anti-mitochondrial antibody for primary biliary cholangitis
- JAK2 mutation for polycythemia vera
- Hepatitis B and C serology
- HIV testing
- Serum protein electrophoresis and immunofixation (myeloma)
- Chest X-ray and CT imaging (lymphoma, solid malignancy)
- Skin biopsy with direct immunofluorescence (dermatitis herpetiformis, autoimmune blistering disease)

- Patch testing (allergic contact dermatitis)

MCQ TRAPS & PITFALLS

Iron deficiency without anemia can cause generalized pruritus — check ferritin.

Polycythemia vera presents with aquagenic pruritus, NOT painful erythromelalgia alone.

Always check LFTs before attributing pruritus to a dermatologic cause — cholestasis is easily missed.

Management of Pruritus

Management should address the underlying cause whenever possible, alongside symptomatic relief for the itch itself.

General Measures

- Emollients and moisturizers: apply liberally and frequently, especially after bathing
- Avoid hot water — use lukewarm water for bathing
- Use soap-free cleansers
- Wear loose-fitting cotton clothing
- Keep fingernails short to minimize excoriation
- Cool compresses for symptomatic relief
- Humidifier use in dry environments

Pharmacologic Treatment

Topical Therapies:

- Emollients: white soft paraffin, aqueous cream — first-line for all pruritus
- Topical corticosteroids: for inflammatory dermatoses (low potency for face/intertriginous areas; high potency for body/extremities)
- Topical calcineurin inhibitors: tacrolimus, pimecrolimus — steroid-sparing for face and flexures
- Menthol 1% or phenol 0.5% in aqueous cream: cooling effect
- Topical capsaicin: for localized neuropathic or uremic pruritus (depletes substance P)
- Coal tar preparations: for psoriasis and chronic eczema

Systemic Therapies:

- First-generation antihistamines (hydroxyzine, chlorpheniramine): sedating — useful for nocturnal itch
- Second-generation antihistamines (cetirizine, loratadine, fexofenadine): non-sedating — for urticaria and histamine-mediated itch
- H2 receptor antagonists (ranitidine, famotidine): added benefit when combined with H1 blockers in urticaria
- Gabapentin and pregabalin: neuropathic pruritus, uremic pruritus, notalgia paresthetica
- SSRIs (paroxetine, sertraline): effective in psychogenic pruritus and paraneoplastic itch

- Dapsone: dermatitis herpetiformis, vasculitic urticaria
- Phototherapy (narrowband UVB): uremic pruritus, prurigo nodularis, HIV-associated pruritus
- Naltrexone (opioid antagonist): cholestatic pruritus
- Cholestyramine: cholestatic pruritus (bile acid sequestrant)

Treatment of Specific Causes

Cholestatic pruritus: Cholestyramine (4g TID) as first-line; rifampicin as second-line; naltrexone as third-line; sertraline as alternative. Refer for liver transplantation evaluation in refractory cases.

Uremic pruritus: Optimize dialysis adequacy; emollients; topical capsaicin; oral gabapentin (100-300mg post-dialysis); phototherapy. Gabapentin dose must be adjusted for renal function.

Hodgkin lymphoma-associated pruritus: Treat the underlying malignancy with appropriate chemotherapy/radiotherapy. Antihistamines and gabapentin may provide symptomatic relief.

Polycythemia vera-associated pruritus: Aspirin (100mg daily), phlebotomy to maintain hematocrit below 45%, cytoreductive therapy (hydroxyurea).

Opioid-induced pruritus: Dose reduction if feasible; switch to alternative opioid; low-dose naloxone infusion; ondansetron.

OSCE PEARLS

Always start with emollients — they are underused and effective for almost all causes of pruritus.

Gabapentin is a safe and effective option for refractory pruritus, but must be dose-adjusted in renal impairment.

Cholestyramine must be taken 2 hours before or after other medications to avoid binding interactions.

Psychogenic Pruritus

Psychogenic pruritus is a diagnosis of exclusion. It should only be considered after thorough investigation has excluded dermatologic, systemic, and neuropathic causes. It is associated with anxiety disorders, depression, obsessive-compulsive disorder, delusions of parasitosis (Ekbom syndrome), and substance misuse (cocaine, amphetamines producing formication — the sensation of insects crawling on the skin).

Patients may present with extensive excoriations that do not conform to a recognizable dermatologic pattern. Delusions of parasitosis is characterized by the fixed, false belief that one is infested with parasites despite lack of evidence. Patients may bring in samples of skin debris (the "matchbox sign"). Management requires a tactful, non-confrontational approach. Psychiatric referral and low-dose antipsychotics (pimozide, risperidone) may be necessary.

IMPORTANT

Never dismiss a patient's symptoms as "just psychological" until organic causes have been thoroughly excluded.

Delusions of parasitosis (Ekbom syndrome) is a fixed false belief — patients will not be reassured by negative investigations. Antipsychotic therapy is the mainstay.

Formication (sensation of insects crawling under the skin) is associated with cocaine and amphetamine use, as well as opioid withdrawal.

High-Yield Summary: Pruritus Without Skin Changes

Red flags that should prompt systemic investigation:

- Pruritus lasting 6 weeks or longer without identifiable skin lesion
- Generalized pruritus with normal skin
- Nocturnal pruritus (consider scabies)
- Aquagenic pruritus (consider polycythemia vera)
- Pruritus worsened by alcohol (consider Hodgkin lymphoma)
- Pruritus with weight loss, night sweats, or fever (B-symptoms)
- New-onset pruritus in an elderly patient (consider malignancy)
- Pruritus on palms and soles (consider cholestasis)

Minimum investigation panel for generalized pruritus without skin changes:

- CBC with differential
- LFTs (including ALP, GGT, bilirubin)
- Renal function tests (BUN, creatinine)
- Thyroid function tests (TSH)
- Fasting glucose / HbA1c
- Iron studies (including ferritin)
- ESR / CRP

Practice Questions

Q1: A 45-year-old woman presents with a 3-month history of generalized pruritus without any rash. She describes intense itching that occurs immediately after taking a warm bath and resolves within 30 minutes. She has noticed facial flushing recently. Her FBC shows hemoglobin 18.2 g/dL, hematocrit 56%, and platelets $520 \times 10^9/L$. What is the most likely diagnosis?

- A) Atopic dermatitis
- B) Hodgkin lymphoma
- C) Polycythemia vera
- D) Primary biliary cholangitis
- E) Chronic kidney disease

Answer: C

Explanation: The key features pointing to polycythemia vera are: (1) **aquagenic pruritus** — itching triggered by contact with warm water, which is highly specific for polycythemia vera; (2) facial flushing (plethora); (3) elevated hemoglobin and hematocrit, and thrombocytosis. JAK2 mutation testing should be performed to confirm the diagnosis. Hodgkin lymphoma causes pruritus that worsens with alcohol, not water contact. Primary biliary cholangitis causes cholestatic pruritus typically localized to palms and soles.

Q2: A 62-year-old man on hemodialysis for end-stage renal disease presents with 6 months of generalized pruritus. His skin appears normal except for scattered excoriations on his back and arms. He has tried multiple emollients and antihistamines without relief. What is the most appropriate next step in management?

- A) Oral prednisolone
- B) Oral gabapentin with dose adjusted for renal function
- C) Cholestyramine

- D) Rifampicin
- E) Patch testing

Answer: B

Explanation: This patient has **uremic pruritus**, diagnosed clinically in the setting of end-stage renal disease with pruritus and no primary skin lesions. After emollients and antihistamines have failed, gabapentin is an effective second-line option for uremic pruritus. It must be dose-adjusted for renal impairment and is typically given post-dialysis (100-300mg). Cholestyramine and rifampicin are treatments for cholestatic pruritus. Patch testing is for allergic contact dermatitis. Oral corticosteroids are not indicated for uremic pruritus.

Q3: A 35-year-old woman presents with intensely pruritic, violaceous, flat-topped papules on her wrists and ankles. On examination, fine white lines are visible on the surface of the papules. Oral examination reveals white, lace-like patches on the buccal mucosa. Which of the following is the most likely finding if a skin biopsy is performed?

- A) IgA deposits at dermal papillae
- B) Saw-tooth pattern of liquefactive degeneration of the basal layer
- C) Intraepidermal blister with acantholysis
- D) Subepidermal blister with eosinophils
- E) Palisading granulomas with central necrobiosis

Answer: B

Explanation: The clinical findings are consistent with **lichen planus**: polygonal, violaceous, flat-topped papules with **Wickham striae** (the fine white lines), affecting the wrists and ankles, with oral mucosal involvement. Histopathology of lichen planus shows a **saw-tooth pattern of liquefactive degeneration of the basal cell layer** with a band-like lymphocytic infiltrate at the dermal-epidermal junction. IgA deposits at dermal papillae are seen in dermatitis herpetiformis. Acantholytic intraepidermal blisters are seen in pemphigus vulgaris. Subepidermal blisters with eosinophils are seen in bullous pemphigoid.

Q4: A 70-year-old man presents with progressive generalized pruritus over the past 4 months. He has lost 6 kg unintentionally and reports drenching night sweats. He has no visible skin lesions except for linear excoriations. Investigations reveal lymphadenopathy on examination, hemoglobin 10.2 g/dL, elevated ESR of 85 mm/hr, and LDH of 580 U/L. Which of the following is the most appropriate next investigation?

- A) Patch testing
- B) Abdominal ultrasound
- C) Excisional lymph node biopsy
- D) Serum protein electrophoresis
- E) Direct immunofluorescence of skin

Answer: C

Explanation: This patient presents with **B-symptoms** (weight loss, night sweats) with generalized pruritus, lymphadenopathy, anemia, elevated ESR, and elevated LDH — all highly suspicious for **Hodgkin lymphoma**. Pruritus occurs in up to 30% of Hodgkin lymphoma patients and may precede diagnosis. The most appropriate next step is **excisional lymph node biopsy** to establish a tissue diagnosis. Reed-Sternberg cells on histology confirm Hodgkin lymphoma. Patch testing and direct immunofluorescence are for dermatologic conditions. Imaging would be part of staging but tissue diagnosis comes first.

Q5: A 55-year-old man presents with intense generalized pruritus without a visible rash. His pruritus is most severe on his palms and soles. On examination, he has mild scleral icterus. Laboratory results show: ALP 420 U/L (elevated), GGT 380 U/L (elevated), total bilirubin 45 µmol/L (elevated), direct bilirubin 35 µmol/L (elevated), AST 85 U/L, ALT 92 U/L. What is the most likely diagnosis?

- A) Uremic pruritus
- B) Cholestatic pruritus due to extrahepatic biliary obstruction
- C) Primary biliary cholangitis

- D) Hodgkin lymphoma
- E) Polycythemia vera

Answer: B

Explanation: This patient has biochemical evidence of **cholestasis** (elevated ALP, GGT, and conjugated bilirubin) with scleral icterus and pruritus on palms and soles. This pattern of obstructive jaundice points to **extrahepatic biliary obstruction** — the most common cause being **pancreatic carcinoma** in a 55-year-old man. An abdominal ultrasound or CT should be performed urgently to identify the site of obstruction. Primary biliary cholangitis typically presents in middle-aged women with positive anti-mitochondrial antibodies and does not cause this degree of conjugated hyperbilirubinemia.

Q6: A 24-year-old healthcare worker presents with a 2-week history of intense pruritus that is markedly worse at night. On examination, thin, serpentine, gray-white tracks are noted in the web spaces between his fingers. Three other family members report similar symptoms. What is the most appropriate treatment?

- A) Oral fluconazole
- B) Topical permethrin 5% cream applied to the patient only
- C) Permethrin 5% cream applied to the patient and all household contacts, wash off after 8-14 hours
- D) Oral ivermectin as a single dose
- E) Topical corticosteroid cream

Answer: C

Explanation: The findings are classic for **scabies**: (1) **burrows** (thin, serpentine, gray-white tracks) in finger web spaces — the pathognomonic lesion; (2) **nocturnal worsening** of itch; (b) **contact history** — multiple family members affected. Treatment is **permethrin 5% cream** applied to the entire body from the neck down, left on for 8-14 hours, and then washed off. **Crucially, all household and close contacts must be treated simultaneously** to prevent reinfection. Oral ivermectin is an alternative for crusted scabies or when topical treatment is impractical.

Q7: A 58-year-old woman with a history of breast cancer treated 2 years ago presents with new-onset generalized pruritus. She has widespread excoriations but no primary skin lesions. Her CBC, LFTs, renal function, and thyroid function are all within normal limits. She reports no change in medications. Which of the following is the most appropriate next step?

- A) Reassure the patient and prescribe emollients only
- B) Patch testing
- C) Age-appropriate cancer recurrence screening and CT imaging
- D) Direct immunofluorescence
- E) Trial of oral antihistamines

Answer: C

Explanation: New-onset generalized pruritus in a patient with a prior history of malignancy should raise suspicion for **cancer recurrence or new primary malignancy** as a paraneoplastic phenomenon, even when initial blood tests are normal. Appropriate cancer screening and imaging (CT chest/abdomen/pelvis or PET-CT depending on the primary tumor type) should be performed before attributing pruritus to a benign cause. Reassurance without further investigation would be inappropriate in this clinical context.